

Dr. Cathaoir Agnew

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SUMMARY

Ph.D. in Artificial Intelligence & Machine Learning, specialising in Computer Vision, with hands-on experience developing real-world AI solutions. In my current role at Akara, I am the sole Computer Vision-focused ML Engineer, building real-time vision systems for healthcare applications, including surgical event detection, temporal action recognition, person detection for UV safety, and OCR and foundation-model-driven workflows for case-cost analysis. I own the full ML lifecycle up to deployment, automate workflows in AWS SageMaker, and collaborate with cross-functional teams to integrate models into production systems.

EXPERIENCE

Machine Learning Engineer

Akara

February 2025 - Present, Dublin

- Built real-time surgical workflow event detection systems for orthopedic and pain management procedures using object detection, multi-object tracking, and temporal video action recognition on thermal video data.
- Developed and trained a person-detection model for a UV disinfection robot, used to trigger safe shutdown of UV emission in the presence of humans.
- Implemented OCR and foundation-model-driven workflows for healthcare case-cost analysis and prediction.
- Own the full ML lifecycle up to deployment, including dataset design, preprocessing, model training, fine-tuning and optimisation for edge devices.
- Improved model robustness with temporal reasoning and post-processing logic to reduce false positives in noisy, real-world operating room environments.
- Automated preprocessing and training workflows using AWS SageMaker to improve scalability.
- Collaborate with cross-functional teams to support production deployment and system integration.
- Continuously research and prototype computer vision models using thermal imaging data for healthcare and robotics applications, driving exploration of scalable real-world deployments.

Computer Vision Researcher

University of Limerick

September 2021 - February 2025, Limerick, Ireland

- Conducted extensive research on the impact of ground truth annotation quality on computer vision performance, resulting in significant insights published in scholarly journals.
- Mentored and supervised summer interns, enhancing their understanding of machine learning principles while fostering a collaborative and supportive research environment.
- Delivered comprehensive instruction on machine learning and deep learning principles, managed computer laboratories and virtual learning platforms and assisted students in identifying and resolving coding issues.
- Published and co-authored seven peer-reviewed publications during my PhD, contributing to the academic knowledge base in the field.

Computer Vision Engineer

University of Limerick

September 2021 - September 2023, Limerick, Ireland

- Collaborated with an industry partner as part of my PhD to conceptualise and develop their first AI system, Vision AI, which automated previously manual processes in container overfill and contamination detection.
- Collaborated with multidisciplinary teams to define data requirements and align project goals, ensuring clear communication to satisfy their needs and expectations throughout the process.
- Trained and deployed the models on Azure Machine Learning Studio, facilitating cloud-based solutions for model deployment and production.
- Achieved 80% mean Average Precision (mAP) on automated overfill container detection and 40% mAP on contamination detection - results published in peer-reviewed journals.
- Applied and researched cutting-edge computer vision techniques to optimise performance, ensuring the final product met both the industry partner's operational needs and quality standards.

SKILLS

Core Skills: Python, Deep Learning, Computer Vision, Statistical Analysis, Mathematics, Artificial Intelligence & Machine Learning, Research & Development, Supervised Learning, Unsupervised Learning, End-to-End ML System Design, Edge and Real-Time ML Deployment, Data Strategy & Dataset Curation

Libraries & Tools: PyTorch, Scikit-Learn, OpenCV, FFMPEG, NumPy, Pandas, Matplotlib, SciPy, Git, AWS, AWS Sagemaker

Techniques: Image Classification, Object Detection, Instance Segmentation, Semantic Segmentation, Feature Engineering, Transfer Learning, Object Tracking, Data Augmentation, Convolution Neural Networks, Transformers, AutoEncoders, GANs, Siamese Neural Networks, Diffusion Models, Pruning, Quantization, Knowledge Distillation

Soft Skills: Analytical Thinking, Problem-Solving, Collaboration, Continuous Learning, Communication, Project Management, Cross-Functional Teamwork

PROJECT

- A collection of my research and side projects can be found on my [website](#). Examples include Object Detection Pseudo-Labelling Pipeline, Enhancing Video Doorbell Footage with Computer Vision, Waste Contamination Detection.

EDUCATION

Ph.D. in Artificial Intelligence & Machine Learning

Minor in Computer Vision • University of Limerick • Ireland • 2024

- Thesis: The Effect of Annotation Quality on Computer Vision in Efficient Waste Management

MSc. in Artificial Intelligence & Machine Learning

University of Limerick • Ireland • 2021

- Thesis: An Investigation into the Timed Up and Go Test Using Inertial Measurements with a Machine Learning Approach

BSc. in Financial Mathematics

University of Limerick • Ireland • 2020

